

Appendix B: Common Taylor Series Expansions

Common Taylor series expansions

#	f	D_f	Expansion	R	D_c
1	e^x	$\mathbb{R}$	$\sum_{n=0}^{+\infty} \frac{x^n}{n!}$	$+\infty$	$\mathbb{R}$
2	$\cos x$	$\mathbb{R}$	$\sum_{n=0}^{+\infty} (-1)^n \frac{x^{2n}}{(2n)!}$	$+\infty$	$\mathbb{R}$
3	$\sin x$	$\mathbb{R}$	$\sum_{n=0}^{+\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$	$+\infty$	$\mathbb{R}$
4	$\cosh x$	$\mathbb{R}$	$\sum_{n=0}^{+\infty} \frac{x^{2n}}{(2n)!}$	$+\infty$	$\mathbb{R}$
5	$\sinh x$	$\mathbb{R}$	$\sum_{n=0}^{+\infty} \frac{x^{2n+1}}{(2n+1)!}$	$+\infty$	$\mathbb{R}$
6	$\frac{1}{1+x}$	$\mathbb{R} \setminus \{-1\}$	$\sum_{n=0}^{+\infty} (-1)^n x^n$	1	$] -1, 1[$
7	$\ln(1+x)$	$] -1, +\infty[$	$\sum_{n=1}^{+\infty} (-1)^{n+1} \frac{x^n}{n}$	1	$] -1, 1]$
8	$\frac{1}{1-x}$	$\mathbb{R} \setminus \{1\}$	$\sum_{n=0}^{+\infty} x^n$	1	$] -1, 1[$
9	$\ln(1-x)$	$] -\infty, 1[$	$-\sum_{n=1}^{+\infty} \frac{x^n}{n}$	1	$[-1, 1[$
10	$\arctan(x)$	$\mathbb{R}$	$\sum_{n=0}^{+\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$	1	$[-1, 1]$
11	$\operatorname{argtanh}(x)$	$] -1, 1[$	$\sum_{n=0}^{+\infty} \frac{x^{2n+1}}{2n+1}$	1	$] -1, 1[$
12	$(1+x)^\alpha$	$] -1, +\infty[$	$1 + \sum_{n=1}^{+\infty} \frac{\alpha(\alpha-1)\dots(\alpha-n+1)}{n!} x^n$		

We recall the identity linking the natural logarithm and the inverse hyperbolic tangent function:

$$\ln\left(\frac{1+x}{1-x}\right) = 2 \operatorname{argtanh}(x),$$

valid for all $x \in] -1, 1[$.